5 A block of mass 27 g is suspended from a spring so that it rests in equilibrium, as shown in Figure 5.1.

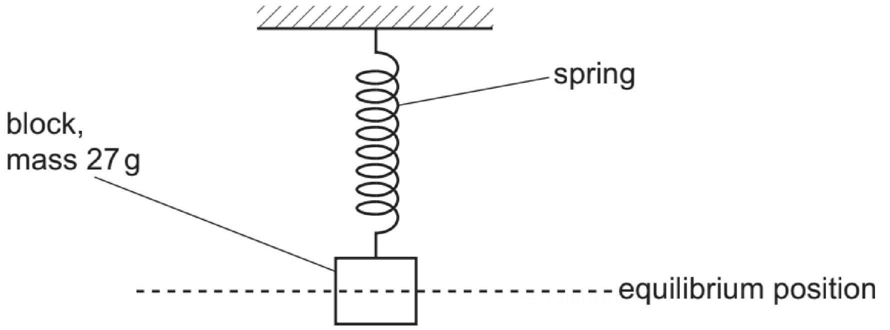

Figure 5.1

The block is displaced downwards through a small distance. The block is released and then oscillates vertically with simple harmonic motion.

The oscillations have a period of 1.7 s and an energy of 13 μJ.

(a) **(i)** Show that the angular frequency of the oscillations is $3.7\,\text{rad}\,\text{s}^{-1}$.

[1]

(ii) State what is meant by the amplitude of the oscillations.

....

....

.... [1]

(iii) Calculate the amplitude of the oscillations.

amplitude = ... m [3]

(b) On Figure 5.2, sketch the variation of the kinetic energy E_K of the block with displacement x from the equilibrium position for one complete oscillation of the block. Add a suitable scale to the x-axis.

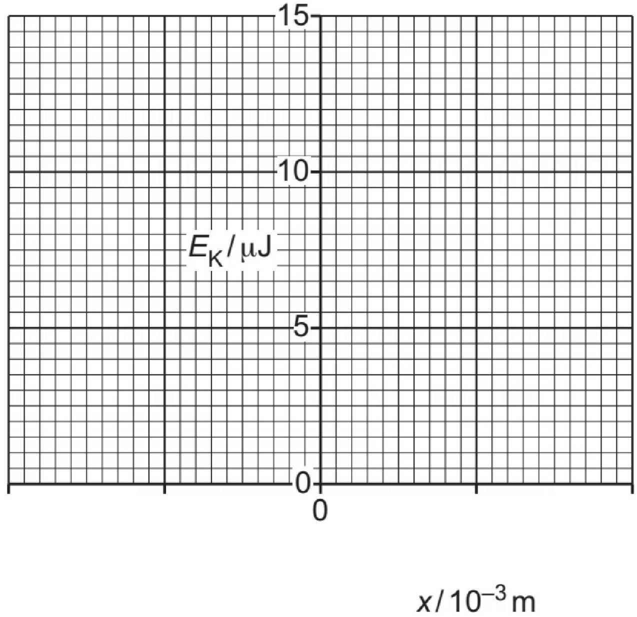

Figure 5.2

[3]

[Total: 8]